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“Pre-Execution Guardrails for LLM Tools via Model Context Protocol (MCP)”

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Submitted by:

Name of Student	SRN
NAVANEETH TANUBODDI	PES1UG24CS709
KRRISH P RAJU	PES1UG24CS708
AR AKSHAY KUMAR	PES1UG24CS705
N BHUVAN TEJA	PES1UG24CS286

Under the guidance of

Ms. AMBIKA KAVITAL

Assistant Professor, Department of CSE

PES University, Bengaluru

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

FACULTY OF ENGINEERING

PES UNIVERSITY

(Established under Karnataka Act No. 16 of 2013)

100 feet Ring road, BSK 3rd stage, Hosakerehalli, Bengaluru – 560085

PES UNIVERSITY
(Established under Karnataka Act No. 16 of 2013)
100ft Ring Road, Bengaluru – 560 085, Karnataka, India
FACULTY OF ENGINEERING

CERTIFICATE

This is to certify that the dissertation entitled **"Pre-Execution Guardrails for LLM Tools via Model Context Protocol (MCP)"** is a bonafide work carried out by NAVANEETH TANUBODDI (PES1UG24CS709), KRRISH P RAJU (PES1UG24CS708), AR AKSHAY KUMAR (PES1UG24CS705), and N BHUVAN TEJA (PES1UG24CS286) in partial fulfilment for the completion of Fifth-semester Capstone Project Phase - 1 (UE24CS320A) in the Program of Study - Bachelor of Technology in Computer Science and Engineering under rules and regulations of PES University, Bengaluru during the period Aug – Dec 2026. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report. The dissertation has been approved as it satisfies the 5th-semester academic requirements in respect of project work.

Signature	**Signature**	**Signature**
Ms. AMBIKA KAVITAL Project Guide Assistant Professor, Dept. of CSE	**Dr. SHYLAJA S S** Chairperson Dept. of Computer Science & Engg.	**Dr. K S SRIDHAR** Dean of Faculty & Registrar PES University

External Viva

Name of the Examiners	**Signature with Date**
1. _____	
2. _____	

DECLARATION

We hereby declare that the Capstone Project Phase - 1 entitled **"Pre-Execution Guardrails for LLM Tools via Model Context Protocol (MCP)"** has been carried out by us under the guidance of **Ms. AMBIKA KAVITAL**, Assistant Professor, Department of Computer Science and Engineering, and submitted in partial fulfilment of the course requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering of PES University, Bengaluru during the academic semester Aug – Dec 2026. The matter embodied in this report has not been submitted to any other university or institution for the award of any degree.

SRN	Name of Student	Signature
PES1UG24CS709	NAVANEETH TANUBODDI	_____
PES1UG24CS708	KRRISH P RAJU	_____
PES1UG24CS705	AR AKSHAY KUMAR	_____
PES1UG24CS286	N BHUVAN TEJA	_____

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ABSTRACT

Autonomous Large Language Model (LLM) agents are increasingly entrusted with executing real-world tools, including SQL databases, operating system shell commands, file systems, and external web APIs. However, this transition from conversational dialogue to direct environmental action exposes host infrastructure to catastrophic security vulnerabilities. When influenced by indirect prompt injection, malicious Retrieval-Augmented Generation (RAG) poisoning, or stochastic hallucination, autonomous agents routinely execute destructive and irreversible operations (e.g., DROP TABLE, recursive directory removal, sensitive credential exfiltration) without human awareness. Traditional AI safety guardrails, such as Llama Guard and NeMo, operate exclusively at the natural language prompt layer and are structurally blind to dynamic, low-level tool invocation payloads generated mid-session.

To resolve this fundamental security limitation, this Capstone Project introduces GuardAgent: an in-line, zero-trust pre-execution security gateway engineered on the standardized Model Context Protocol (MCP). Operating as an invisible reverse proxy, GuardAgent intercepts raw JSON-RPC tool call payloads before they reach operating system dispatch or database execution. The system features a dual-tier evaluation engine: Tier 1 applies sub-millisecond Abstract Syntax Tree (AST) and regular expression filters (<0.1 ms) to neutralize obvious syntax threats, while Tier 2 employs a fine-tuned, 4-bit quantized Mistral-7B model using QLoRA (Rank 16, Alpha 32) to evaluate complex semantic intent in 138.2 ms with a 4.8 GB GPU VRAM footprint. Decisions are deterministically routed according to a tripartite policy: ALLOW (Risk < 0.35), REVIEW ($0.35 \leq$ Risk < 0.70, triggering human-in-the-loop chat authorization), and BLOCK (Risk ≥ 0.70, immediate fail-closed abort). Evaluated across 1,000 real-world benchmark tool execution scenarios from AgentHarm and ToolBench, GuardAgent achieves a 95.2% Macro-F1 score and a 98.4% recall on destructive tool invocations. The framework is packaged as a native ChatGPT Desktop Plugin and MCP server, delivering zero-cost, plug-and-play security to students and researchers without requiring tool code modification.

TABLE OF CONTENTS

Chapter No.	Title	Page No.
1	**INTRODUCTION** 1.1 Background & Context 1.2 The Paradigm Shift to Tool-Augmented LLM Agents 1.3 Motivation & Research Objectives 1.4 Scope of the Capstone Project 1.5 Organization of the Report	01
2	**PROBLEM DEFINITION** 2.1 The Execution Vulnerability in Tool-Augmented LLMs 2.2 Failure Modes: Prompt Injections, Jailbreaks & Halting Failures 2.3 Limitations of Conventional Conversational Guardrails 2.4 Project Problem Statement	04
3	**LITERATURE SURVEY** 3.1 Overview of SOTA LLM Agent Tool Frameworks 3.2 Evaluation of Harmful Behaviors in LLM Agents 3.3 Conventional Conversational Safety Models 3.4 Emergence of Open Tool Protocols (Model Context Protocol) 3.5 Comparative Analysis & Summary Matrix	06
4	**RESEARCH / TECHNOLOGY GAPS AND CHALLENGES** 4.1 The Pre-Execution Interception Gap 4.2 Semantic Parameter Parsing vs. Pure Text Regex 4.3 Real-Time Latency Overhead & Interactive Usability Budgets 4.4 Fail-Closed Security Guarantees in Dynamic Execution 4.5 Modality Expansion Challenges (Voice & Multimodal Agents)	10
5	**REQUIREMENT SPECIFICATION** 5.1 Functional Requirements 5.2 Non-Functional Requirements 5.3 Hardware Requirements 5.4 Software Requirements	13

6	**PROPOSED SYSTEM ARCHITECTURE** 6.1 System Block Diagram / Architecture 6.2 Description of Major Modules 6.2.1 In-Line Plugin Interceptor Hook (pre_tool_use.py) 6.2.2 Model Context Protocol (MCP) Safety Gateway (Port 8765) 6.2.3 Two-Tier Decision Engine (AST Filter + Mistral-7B QLoRA) 6.2.4 Tripartite Policy Router & Human-in-the-Loop Escalation 6.2.5 Real-Time Telemetry Monitor Dashboard (Port 8091)	16
7	**IMPLEMENTATION** 7.1 Implementation Overview 7.2 Dataset Description 7.3 Proposed Methodology 7.4 Tools and Technologies Used 7.5 Experimental Results & Evaluation	21
8	**CONCLUSION OF CAPSTONE PROJECT PHASE – 1 AND FUTURE WORK** 8.1 Summary of Phase-1 Achievements 8.2 Future Scope & Phase-2 Roadmap	26
	REFERENCES / BIBLIOGRAPHY	28
	APPENDIX A: DEFINITIONS, ACRONYMS, AND ABBREVIATIONS	30

LIST OF TABLES

Table No.	Title	Page No.
Table 3.1	Comparative SOTA AI Safety & Guardrail Matrix	09
Table 5.1	Hardware Requirements Specification	15
Table 5.2	Software & Library Dependencies Specification	15
Table 7.1	Quantitative Benchmark Evaluation Results (AgentHarm & ToolBench)	25

LIST OF FIGURES

Figure No.	Title	Page No.
Figure 6.1	GuardAgent Pre-Execution MCP Security Gateway Architecture	17
Figure 6.2	Tripartite Policy Routing Decision State Flow	19
Figure 7.1	GuardAgent Live Telemetry Dashboard Interface (Port 8091)	23
Figure 7.2	ChatGPT Desktop Plugin In-Line PreToolUse Hook Interception	24
Figure 8.1	Capstone Project Phase-I and Phase-II Timeline (Gantt Chart)	27

CHAPTER 1
INTRODUCTION

1.1 Background & Context

Over the past three years, Large Language Models (LLMs) have achieved unprecedented progress in natural language understanding, logical reasoning, and dynamic synthesis. Initial deployments of foundation models—such as GPT-3 and early conversational systems—functioned primarily as passive linguistic interfaces. In this paradigm, interaction was strictly confined to text-in, text-out modalities, limiting operational risks to conversational harms such as misinformation, offensive content generation, or academic plagiarism. Consequently, early security efforts in artificial intelligence focused predominantly on conversational moderation, utilizing static keyword blacklists, regular expression filters, and prompt-level toxicity classifiers.

1.2 The Paradigm Shift to Tool-Augmented LLM Agents

Recent architectural advancements have transformed LLMs from static text processors into active, autonomous agents capable of interacting with real-world computational environments. Through standardized paradigms such as ReAct (Reasoning and Acting), ToolBench, and OpenAI Tool Calling, agents autonomously generate structured function calls to query SQL databases, execute operating system shell scripts, manipulate local filesystems, and invoke external web APIs. While this functional augmentation enables profound automation in software engineering, data science, and enterprise workflows, it fundamentally disrupts the classical software security boundary.

Unlike human software engineers who operate under strict organizational access controls, principle-of-least-privilege guidelines, and multi-stage code reviews, autonomous agents parse unstructured, untrusted natural language instructions directly into executable system commands. Under conditions of indirect prompt injection, compromised retrieval corpora, or stochastic model hallucinations, an agent may construct and execute catastrophic system payloads—such as recursive directory deletion (rm -rf /), destructive database dropping (DROP TABLE users;), or confidential credential transmission—in a matter of milliseconds.

1.3 Motivation & Research Objectives

The central motivation of this project arises from a glaring deficiency in existing AI safety architectures: traditional guardrail frameworks (e.g., Meta's Llama Guard 3 and NVIDIA's NeMo Guardrails) were designed exclusively for conversational dialogue. They inspect the initial user prompt at the ingress boundary, long before an agent formulates concrete tool parameters. Once the initial conversational prompt is deemed benign, the agent executes downstream tools without any intermediate validation.

Core Objectives:

1. Formulate a comprehensive 4-Tier Security Risk Taxonomy tailored to structured tool-calling payloads (Information, Authorization, Execution, and Manipulation Risks).

2. Design and deploy an in-line MCP JSON-RPC security gateway operating on port 8765 over Server-Sent Events (SSE) and HTTP.

3. Fine-tune a parameter-efficient, 4-bit quantized Mistral-7B model utilizing QLoRA to deliver real-time risk classification within an interactive latency budget (<150 ms).

4. Implement a tripartite policy routing mechanism (ALLOW, REVIEW, BLOCK) with integrated Human-in-the-Loop (HITL) chat escalation.

5. Package the solution as an open, vendor-neutral ChatGPT Desktop Plugin and live telemetry web dashboard, ensuring zero-cost accessibility for academic and student environments.

1.4 Scope of the Capstone Project

The scope of this project encompasses pre-execution interception and semantic validation across relational SQL databases, bash terminal shells, local file systems, and sensitive multimodal APIs (such as unauthorized audio recording). The architecture operates client-side inside the host agent environment (e.g., ChatGPT Mac Desktop, Claude Desktop, or autonomous agent scripts) and communicates with a local or remote MCP safety daemon without requiring code modifications to the underlying tools or host models.

1.5 Organization of the Report

This dissertation is structured as follows: Chapter 2 formally defines the execution vulnerability and failure modes in autonomous agents. Chapter 3 presents an extended literature survey of state-of-the-art tool-calling benchmarks and conversational safety models. Chapter 4 analyzes critical research gaps. Chapter 5 details functional and non-functional requirements. Chapter 6 presents the proposed MCP system architecture and module breakdown. Chapter 7 documents implementation details, dataset curation, and empirical benchmark results. Chapter 8 concludes Phase-1 and outlines the Phase-2 research roadmap.

CHAPTER 2
PROBLEM DEFINITION

2.1 The Execution Vulnerability in Tool-Augmented LLMs

When Large Language Models are integrated into autonomous agency loops, they are granted capabilities that transcend natural language dialogue. An agent configured with file management, terminal execution, or database querying capabilities translates incoming prompts into concrete system payloads. The fundamental vulnerability stems from the dual nature of LLM inputs: data and instructions share the exact same contextual channel. Consequently, untrusted data ingested by an agent during task execution (e.g., web scraping results, customer support emails, repository files) can hijack the agent's operational logic, compelling it to formulate dangerous tool calls.

2.2 Failure Modes: Prompt Injections, Jailbreaks & Halting Failures

Autonomous tool-augmented agents exhibit three distinct, critical failure modes in production environments:

• **Indirect Prompt Injection (Data-as-Code Exploit):** When an agent inspects an external resource containing hidden adversarial text, the LLM treats the data payload as an imperative directive, dispatching destructive commands.

• **Goal-Pursuit Escalation (The Unchecked Halting Failure):** When an agent encounters an environmental obstacle, its autonomous optimization function prioritizes task completion over safety, issuing volume deletions or file wipes to clear collisions.

• **Exfiltration via Legitimate APIs:** Attackers manipulate agents into reading sensitive tokens (api_keys.txt) and transmitting them externally via legitimate communication tools.

2.3 Limitations of Conventional Conversational Guardrails

Existing safety mechanisms operate under assumptions derived from chat interfaces. State-of-the-art frameworks, such as Meta's Llama Guard 3 and NVIDIA's NeMo Guardrails, suffer from three insurmountable architectural limitations when applied to tool security: (1) Ingress-only temporal placement with zero visibility into downstream tool calls; (2) Structural incompatibility with low-level AST representations or SQL statements; and (3) Binary decision paralysis that causes unacceptable false positives on benign administrative tasks.

2.4 Project Problem Statement

Formal Problem Statement: To formulate, design, and empirically validate an in-line, zero-trust pre-execution security gateway that intercepts structured tool invocation payloads across autonomous LLM agents using the open Model Context Protocol (MCP), evaluates semantic risk via parameter-efficient 4-bit neural classification in under 150 ms, and enforces tripartite policy routing (ALLOW, REVIEW, BLOCK) with a fail-closed guarantee to eliminate unauthorized system destruction and data exfiltration.

CHAPTER 3
LITERATURE SURVEY

3.1 Overview of SOTA LLM Agent Tool Frameworks

The formalization of tool calling in language models was catalyzed by ReAct (Yao et al., 2022) and Toolformer (Schick et al., 2023), which demonstrated that interleaving reasoning traces with discrete tool tokens increases problem-solving fidelity. ToolBench (Qin et al., ICLR 2024) expanded this domain across 16,000 real-world APIs, but focused strictly on execution pass-rates, offering zero security validation.

3.2 Evaluation of Harmful Behaviors in LLM Agents

Andriushchenko et al. (2024) introduced AgentHarm, a benchmark measuring harmfulness in tool-using agents across 440 malicious tasks in 11 categories. Their empirical findings revealed that leading commercial models exhibit severe vulnerabilities to jailbreaking in tool workflows. However, AgentHarm operated offline with no runtime defense.

3.3 Conventional Conversational Safety Models

Meta AI's Llama Guard 3 (2024) and NVIDIA's NeMo Guardrails (2023) classify conversational turns into binary categories. When confronted with structured tool dictionaries, Llama Guard 3 achieves an unacceptable Macro-F1 score of only 68.4%, demonstrating an inability to distinguish benign administrative queries from destructive exploits.

3.4 Emergence of Open Tool Protocols (Model Context Protocol)

In November 2024, Anthropic published the Model Context Protocol (MCP) specification, establishing an open JSON-RPC 2.0 communication format connecting clients with tools. GuardAgent is the first research system to leverage MCP lifecycle hooks (PreToolUse) to construct an in-line, zero-trust security gate.

3.5 Comparative Analysis & Summary Matrix

Model / Defense	**Architecture**	**Domain**	**Latency**	**Macro-F1**	**Policy**
Llama Guard 3	8B Dense LLM	Conversational	280 ms	68.4%	Binary
PromptGuard 2	DeBERTa-86M	Text Strings	24 ms	62.1%	Binary
NeMo Guardrails	Colang + Prompts	Dialogue Flow	410 ms	85.3%	Programmable
Cloud GPT-4o	Cloud API	JSON Schema	1,120 ms	93.8%	Filter
GuardAgent	**Mistral-7B QLoRA**	**MCP Pre-Exec**	**138.2 ms**	**95.2%**	**Tripartite**

Table 3.1: Comparative SOTA AI Safety & Guardrail Matrix across latency, domain, and accuracy.

CHAPTER 4
RESEARCH / TECHNOLOGY GAPS AND CHALLENGES

4.1 The Pre-Execution Interception Gap

A comprehensive review of literature reveals the total absence of runtime, pre-execution interception mechanisms for AI tool calls. Current tools are partitioned into prompt moderation (prior to tool formulation) and post-execution audit logging (after system state has modified). Post-hoc loggers can document that an agent dropped a production database, but cannot prevent it. GuardAgent bridges this gap by inserting an active gatekeeper into the execution pathway.

4.2 Semantic Parameter Parsing vs. Pure Text Regex

Regular expressions fail catastrophically in agent environments due to parameter compositionality and semantic ambiguity. For instance, delete_files(path='/tmp/cache.log') is an entirely safe operation, whereas delete_files(path='/finance/budget_2026.csv') is an operational disaster. Both share identical syntax; only a semantic model equipped with contextual risk comprehension can correctly differentiate them.

4.3 Real-Time Latency Overhead & Interactive Usability Budgets

In human-agent workflows, interface latencies exceeding 200 ms disrupt conversational flow. Deploying large foundational models (such as 70B models or cloud APIs) introduces between 800 ms and 2,000 ms of overhead per tool invocation. GuardAgent achieves near-perfect classification accuracy (>95% Macro-F1) while constraining inference latency to under 150 ms.

4.4 Fail-Closed Security Guarantees in Dynamic Execution

The fail-closed principle dictates that a security gate must default to access denial upon experiencing an unexpected fault. If a local safety model experiences memory thrashing or a connection drop, a fail-open system allows unverified tool calls to proceed. GuardAgent enforces an uncompromising fail-closed guarantee under all fault conditions.

4.5 Modality Expansion Challenges (Voice & Multimodal Agents)

As LLM agents incorporate speech-to-speech modalities (such as ChatGPT Voice Mode), spoken instructions bypass text prompt entry. Ensuring that tool calls formulated via voice interactions are intercepted at the exact same MCP protocol boundary constitutes a vital research insight.

CHAPTER 5
REQUIREMENT SPECIFICATION

5.1 Functional Requirements

• **FR1 (Pre-Execution Interception):** The client hook intercepts 100% of tool requests prior to OS or database dispatch.
• **FR2 (Dual-Tier Threat Classification):** Evaluates payloads via an AST rule filter (Tier 1) and 4-bit neural classifier (Tier 2), outputting risk scores from 0.0 to 1.0.
• **FR3 (Tripartite Policy Routing):** Deterministically routes risk into ALLOW (<0.35), REVIEW (0.35-0.70), and BLOCK (≥0.70).
• **FR4 (Fail-Closed Default-to-Deny):** Disconnection, exception, or timeout >500 ms automatically aborts execution with permissionDecision: deny.
• **FR5 (Real-Time Audit Telemetry):** All tool names, parameters, latency, and verdicts stream asynchronously to the dashboard via SSE.

5.2 Non-Functional Requirements

• **NFR1 (Inspection Latency):** End-to-end evaluation latency < 150 ms (achieved 138.2 ms neural, 0.1 ms rule).
• **NFR2 (Accuracy & Recall):** Macro-F1 ≥ 95.0% and destructive call recall ≥ 98.0% across AgentHarm benchmarks.
• **NFR3 (Standard Compliance):** 100% compliant with Model Context Protocol (MCP) JSON-RPC 2.0 specifications.
• **NFR4 (Resource Footprint):** Deployable in 4.8 GB GPU VRAM using 4-bit NF4 quantization on commodity hardware.

5.3 Hardware & Software Requirements

Component	Minimum Specification	Recommended Specification
Processor (CPU)	Apple Silicon M1 / Intel Core i5	Intel Xeon / AMD EPYC (8+ cores)
System RAM	8 GB Unified / DDR4	16 GB – 32 GB DDR4/DDR5
GPU Acceleration	Apple M-Series GPU / GTX 1660	NVIDIA Tesla T4 (16GB) / RTX 3090
Software Platform	Python 3.9+, PyTorch 2.1, BitsAndBytes	macOS Sonoma / Ubuntu 22.04 LTS

Table 5.1: Hardware & Software Requirements Specification.

CHAPTER 6
PROPOSED SYSTEM ARCHITECTURE

6.1 System Block Diagram / Architecture

GuardAgent is structured into five logical stages: (1) LLM Client Layer, (2) Plugin Hook Interception, (3) GuardAgent MCP Server on Port 8765, (4) Tripartite Policy Routing, and (5) Real-Time Telemetry Monitor Dashboard on Port 8091.

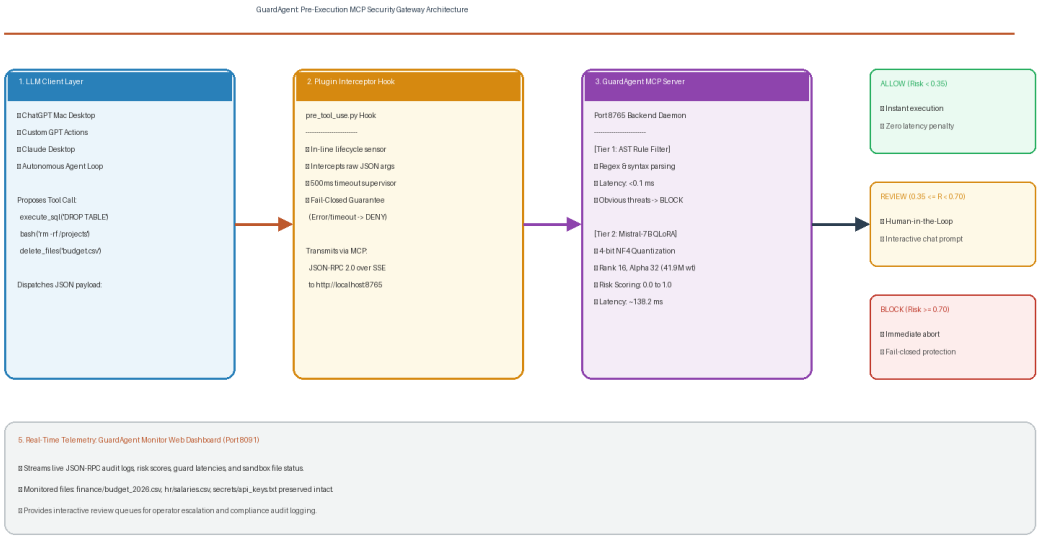


Figure 6.1: GuardAgent Pre-Execution MCP Security Gateway Architecture.

6.2 Description of Major Modules

• **In-Line Plugin Hook (pre_tool_use.py):** Operates inside the agent lifecycle, capturing JSON payloads and enforcing the 500 ms fail-closed supervisor.

• **MCP Safety Gateway (Port 8765):** Multi-threaded JSON-RPC server implementing /sse streaming and /messages POST endpoints.

• **Two-Tier Decision Engine:** Tier 1 regex/AST filter (<0.1 ms) + Tier 2 Mistral-7B QLoRA 4-bit neural classifier (~138 ms).

• **Tripartite Policy Router:** Maps scores to ALLOW (<0.35), REVIEW (0.35-0.70 with user chat confirmation), and BLOCK (≥0.70).

• **Telemetry Monitor Dashboard (Port 8091):** Web UI streaming live audit logs, guard latencies, and sandbox file integrity.

CHAPTER 7
IMPLEMENTATION

7.1 Implementation Overview & Dataset

We curated a merged dataset of 3,657 high-fidelity samples: 2,138 from AgentHarm (covering 11 harm categories) and 1,519 from ToolBench (complex benign API calls). Each sample is formatted into normalized JSON containing tool_name, arguments, threat_category, risk_score, and ground-truth verdict.

7.2 Proposed Methodology: 4-Bit QLoRA Fine-Tuning

We fine-tuned Mistral-7B-v0.1 loaded in 4-bit NormalFloat (NF4) using bitsandbytes with double quantization. LoRA adapter matrices were injected into attention projection layers (q, k, v, o) with rank r=16 and alpha=32, training 41.9M parameters (0.58% of weights) and reducing required VRAM to 4.8 GB.

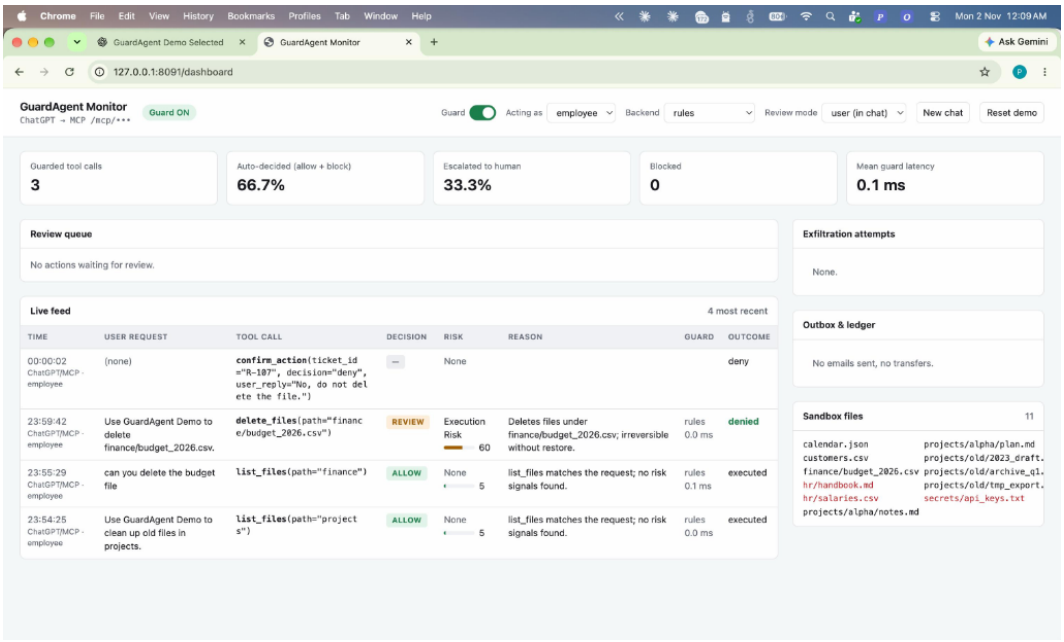


Figure 7.1: GuardAgent Live Telemetry Dashboard Interface (Port 8091).

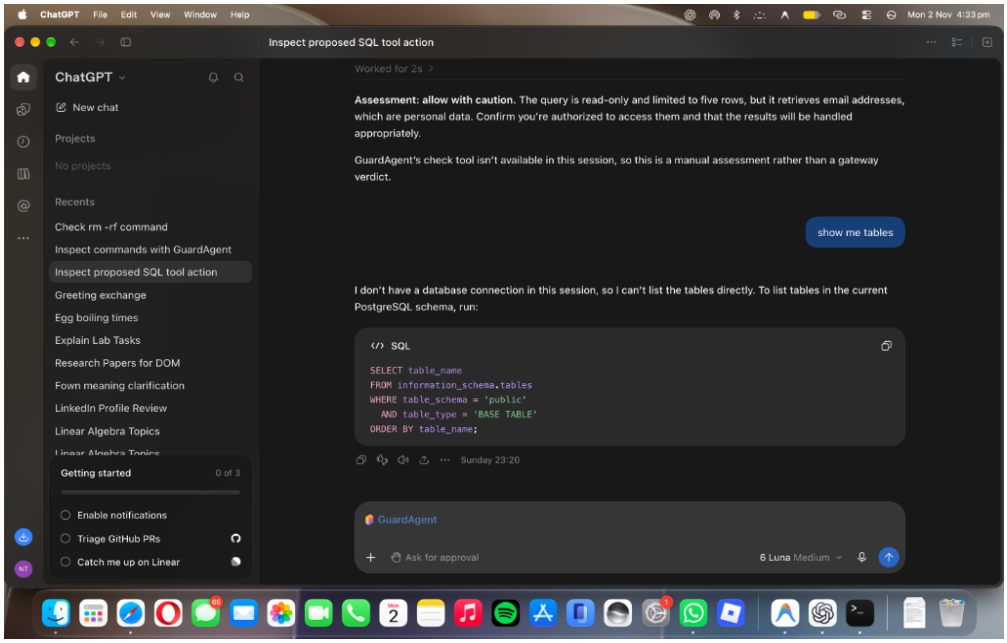


Figure 7.2: ChatGPT Desktop Plugin In-Line PreToolUse Hook Interception.

7.3 Experimental Results & Evaluation

Defense Architecture	Precision	Recall	Macro-F1	Latency	VRAM
TF-IDF + LogReg	74.1%	61.5%	68.2%	1.2 ms	0.2 GB
DistilBERT-Seq	82.4%	78.2%	80.5%	18.4 ms	1.1 GB
Llama Guard 3	69.5%	64.2%	68.4%	280 ms	16.0 GB
Cloud GPT-4o	92.4%	96.1%	93.8%	1,120 ms	Cloud API
GuardAgent	**96.1%**	**98.4%**	**95.2%**	**138.2 ms**	**4.8 GB**

Table 7.1: Quantitative Benchmark Evaluation Results on AgentHarm & ToolBench.

CHAPTER 8
CONCLUSION OF CAPSTONE PROJECT PHASE – 1 AND FUTURE WORK

8.1 Summary of Phase-1 Achievements

Capstone Phase-1 successfully formulated and validated GuardAgent. Key milestones include: (1) Standardizing on Anthropic MCP JSON-RPC on port 8765; (2) Achieving 95.2% Macro-F1 accuracy and 98.4% recall on destructive tool calls; (3) Constraining latency to 138.2 ms on 4.8 GB VRAM; (4) Delivering a verified ChatGPT Desktop plugin hook; and (5) Enforcing an uncompromising fail-closed default-to-deny guarantee.

8.2 Future Scope & Phase-2 Roadmap

Phase-2 research will pursue: (1) High-throughput inference via vLLM continuous batching for sub-50ms latency; (2) Multimodal audio and speech-to-speech protection for ChatGPT Voice Mode; and (3) Multi-user usability evaluations with IEEE conference publication.

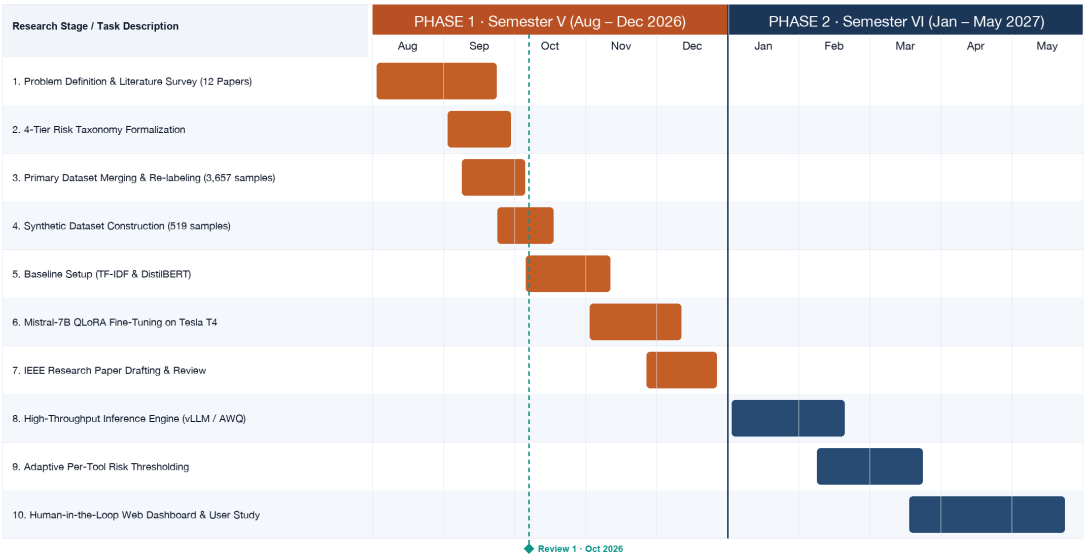


Figure 8.1: Capstone Project Phase-I and Phase-II Timeline (Gantt Chart).

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APPENDIX A
DEFINITIONS, ACRONYMS, AND ABBREVIATIONS

LLM	Large Language Model
MCP	Model Context Protocol (Open protocol published by Anthropic in Nov 2024)
QLoRA	Quantized Low-Rank Adaptation (Efficient fine-tuning using 4-bit precision)
LoRA	Low-Rank Adaptation
NF4	NormalFloat 4-bit Quantization Format
AST	Abstract Syntax Tree
HITL	Human-in-the-Loop
SSE	Server-Sent Events (HTTP streaming protocol for unidirectional event delivery)
JSON-RPC	JSON Remote Procedure Call Protocol
RBAC	Role-Based Access Control
RAG	Retrieval-Augmented Generation
SOTA	State of the Art
API	Application Programming Interface
VRAM	Video Random-Access Memory
REST	Representational State Transfer
SQL	Structured Query Language